

Solutions

Question 1

$$a) \quad R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

$$\cdot \quad t=2; \quad R_2 = \frac{108 - 100}{100} = 0.08 \text{ (8\%)}$$

$$\cdot \quad t=3;$$

$$R_3 = \frac{103 - 108}{108} = -0.046 \text{ (or -4.63\%)}$$

$$\cdot \quad t=4;$$

$$R_4 = \frac{115 - 103}{103} = 0.1165 \text{ (or 11.65\%)}$$

$$\cdot \quad t=5$$

$$R_5 = \frac{110 - 115}{115} = -0.043 \text{ (or -4.35\%)}$$

b)

$$r_2 = \ln(108/100) = 0.076 \text{ (7.7\%)}$$

$$r_3 = \ln(103/108) = -0.047 \text{ (-4.74\%)}$$

$$r_4 = \ln(115/103) = 0.11 \text{ (11.2\%)}$$

$$r_5 = \ln \left(110/115 \right) = -0.044 \text{ (}-4.45\%)$$

c) Verification & accuracy of the approximation
 $r_t \approx R_t$

Absolute difference $(|R_t - r_t|) \rightarrow A.D.$

$$\textcircled{1} \quad AD = 0.003039$$

$$\textcircled{2} \quad AD = 0.001106$$

$$\textcircled{4} \quad AD = 0.006802$$

$$\textcircled{5} \quad AD = 0.000974$$

for $t=4$, it is least accurate.

Reason
$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right) = \ln (1 + R_t) \approx R_t$$

for small value of simple return.

at $t=4$, asset experienced its largest price,
 meaning it cause it behaviour

... ..

more demand from linear version one.

Q 2

a) we know that

$$E[ax + bY] = aE[X] + bE[Y];$$

$$R_p = w_1 R_1 + w_2 R_2$$

$$\begin{aligned} E[R_p] &= E[w_1 R_1 + w_2 R_2] \\ &= w_1 E[R_1] + w_2 E[R_2] \end{aligned}$$

$$E[R_p] = 0.4 \times 0.15 + 0.6 \times 0.07$$

$$E[R_p] = 0.102 \text{ or } 10.2\%$$

b) correlation ($\rho_{1,2}$) heavily dictates the portfolios total risk, it has no impact on expected return.

$$E[X+Y] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x+y) f_{X,Y}(x,y) dx dy.$$

integral operates linearly so

$$E[X+Y] = \int x f_X(x) dx + \int y f_Y(y) dy$$

$$\Rightarrow E[X] + E[Y]$$

Q.3

$$R_p = w_1 R_1 + w_2 R_2$$

$$\sigma_p^2 = \text{Var}(w_1 R_1 + w_2 R_2)$$

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{COV}(X, Y)$$

$$\text{Var}(w_1 R_1 + w_2 R_2) = \text{Var}(w_1 R_1) + \text{Var}(w_2 R_2) + 2 \text{COV}(w_1 R_1, w_2 R_2)$$

$$\bullet \text{Var}(ax) = a^2 \text{Var}(x).$$

$$\Rightarrow w_1^2 \text{Var}(R_1) + w_2^2 \text{Var}(R_2) + 2 w_1 w_2 \text{COV}(R_1, R_2)$$

$$\sigma_p^2 = w_1^2 \sigma_1 + w_2^2 \sigma_2 + 2 w_1 w_2 \text{COV}(R_1, R_2)$$

$$\rho_{12} = \frac{\text{COV}(R_1, R_2)}{\sigma_1 \cdot \sigma_2}$$

Q.4

$$a) \sigma_p \text{ for } \rho = 1, 0, -1;$$

$$\sigma_{p_1}^2 = 0.0156 + 0.0036 + 0.015(1)$$

$$\sigma_{p_1} = 0.185 \text{ (18.5\%)}$$

$$\sigma_{p_0}^2 = 0.0156 + 0.0036$$

$$\sigma_{p_0} = 0.1387 \text{ (13.87\%)}$$

$$\sigma_{p_{-1}}^2 = 0.0156 + 0.0036 - 0.015$$

$$\sigma_{p_{-1}} = 0.065 \text{ (6.5\%)}$$

b). when $\rho = 1$

the σ_p equals to the weighted average of individual risks.

$$\sigma_p^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B (1) \sigma_A \sigma_B$$

$$\sigma_p^2 = (w_A \sigma_A + w_B \sigma_B)^2$$

$$\boxed{\sigma_p = |w_A \sigma_A + w_B \sigma_B|}$$

c) for $\rho = -1$

$$\sigma_p^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 - 2w_A w_B \sigma_A \sigma_B$$

$$\sigma_p^2 = (w_A \sigma_A - w_B \sigma_B)^2$$

$$\sigma_p = |w_A \sigma_A - w_B \sigma_B|$$

for risk to be zero $\sigma_p = 0$

$$w_A \sigma_A = w_B \sigma_B$$

$$w_A \sigma_A = (1 - w_A) \sigma_B$$

$$w_A^* = \frac{\sigma_B}{\sigma_A + \sigma_B} = \frac{0.12}{0.25 + 0.12} = \frac{0.3243}{\text{or } \boxed{32.43} \%}$$

Ques. 5

When we combine 2 assets into a portfolio the overall risk isn't simply a straight line average of their individual risks, unless the assets move perfectly in $\rho = 1$.

Whenever $\rho < 1$ a mathematical phenomenon

called diversification occurs, pulling the portfolio's total risks below that weighted average.

Q.6

a) Construct a 3×3 matrix.

$$\Sigma_{ij} = \rho_{ij} \sigma_i \sigma_j$$

$$\Sigma_{11} = \sigma_1^2 = 0.0625$$

$$\Sigma_{22} = \sigma_2^2 = 0.0144$$

$$\Sigma_{33} = \sigma_3^2 = 0.0016$$

$$\Sigma_{12} = \rho_{12} \sigma_1 \sigma_2 = 0.0120 \quad \Sigma_{23} = 0.00096$$

$$\Sigma_{13} = 0.0010$$

$$\Sigma = \begin{pmatrix} 0.0625 & 0.0120 & 0.0010 \\ 0.0120 & 0.0144 & 0.00096 \\ 0.0010 & 0.00096 & 0.0016 \end{pmatrix}$$

b) $\Sigma_{12} = \Sigma_{21}$ } from this observation
 that

$$\left. \begin{aligned} \Sigma_{13} &= \Sigma_{31} \\ \Sigma_{23} &= \Sigma_{32} \end{aligned} \right\} \begin{array}{l} \text{we can say then} \\ \text{it is symmetric} \end{array}$$

Q. 7.

$$a). \text{ Sharpe ratio} = \frac{E[R_p] - R_f}{\sigma_p}$$

Portfolio X, Y & Z

$$SR_X = \frac{0.14 - 0.04}{0.22} = 0.4545$$

$$SR_Y = \frac{0.09 - 0.04}{0.10} = .5$$

$$SR_Z = \frac{0.2 - 0.04}{0.35} = 0.4571$$

b) Rank

$$Y > Z > X.$$

Preferred $\rightarrow Y$

reason $\rightarrow$ S.R. represents slope of the capital allocation line connecting risk free asset

c) reason is given in b part.

Q 8

a) GMV $\rightarrow$ risk minimizing choice for an investor who must allocate 100% of their wealth across a given set of risky assets without borrowing or lending at a risk free rate

X-axis $\rightarrow$ risk

Y-axis $\rightarrow$ expected return;

on moving left risk reduces, since GMV is minimum risk so it is the leftmost point

b) argument is incorrect.

upper half $\rightarrow$ maximise expected return

c) CML $\rightarrow$ a risk free asset into an

economy consisting of entire market universe

→ Anchor point → $\sigma = 0$ & $E[R] = R_f$

→ Slope of CML is a Sharpe ratio.

Q. 9.

$\sigma = 30\%$ Annual

a) $\hat{\mu}_{\text{annual}} \rightarrow \frac{\sigma_{\text{ann}}}{\sqrt{T}} = \frac{30\%}{\sqrt{3}} = 17.32\%$

$$\sigma_{\text{monthly}} = \frac{30\%}{\sqrt{12}} = 8.66\%$$

$$\hat{\mu}_{\text{monthly}} = \frac{\sigma_{\text{mon.}}}{\sqrt{N}} = \frac{8.66}{\sqrt{36}} = 1.44\%$$

$$\text{Annual Stan. error} = 1.44\% \times 12 \approx 17.32\%$$

$$\sigma = 30\% (\text{Monthly})$$

$$\mu_{\text{monthly}} = \frac{\sigma_{\text{mon.}}}{\sqrt{N}} = \frac{30\%}{\sqrt{36}} = 5\%$$

$$\text{annual stand. error} = 5 \times 12 = 60\%$$

we will rely on $\sigma = 30\%$ Annual.

U
b) MRO processes noisy inputs, it actually acts as an error maximizer. The optimizer combs through the assets universe & aggressively assigns large positive weight to assets that happens to have large, +ive estimation errors & it assign large -ive weights or completely drops assets that suffered from +ive esti. error.

c) $N = 50$ assets

$$\text{unique} = \frac{N(N+1)}{2}$$

$$\text{unique parameter} = 1275$$

When N is large relative to sample size, the small eigen value of sample covariance matrix are systematically underestimated, while large eigen value are overestimated.

